

UNIT-IVPERIPHERAL INTERFACING8255:-

→ 8255 Programmable peripheral Interface (PPI) is very popular & versatile I/O chip that can be easily programmed to function in several different configuration.

Features:-

→ 3 8-bit TTL-compatible registers (or) Ports, which allow programmers to control digital O/Ps, I/Ps, (or) a combination of both.

i) It has 3 8-bit ports A, B, C connected to the O/P pins.

ii) Port C is divided into 2 groups, port C - upper [PCU] & port C Lower [PCL], of 4 bits each. Each of them can be programmed independently.

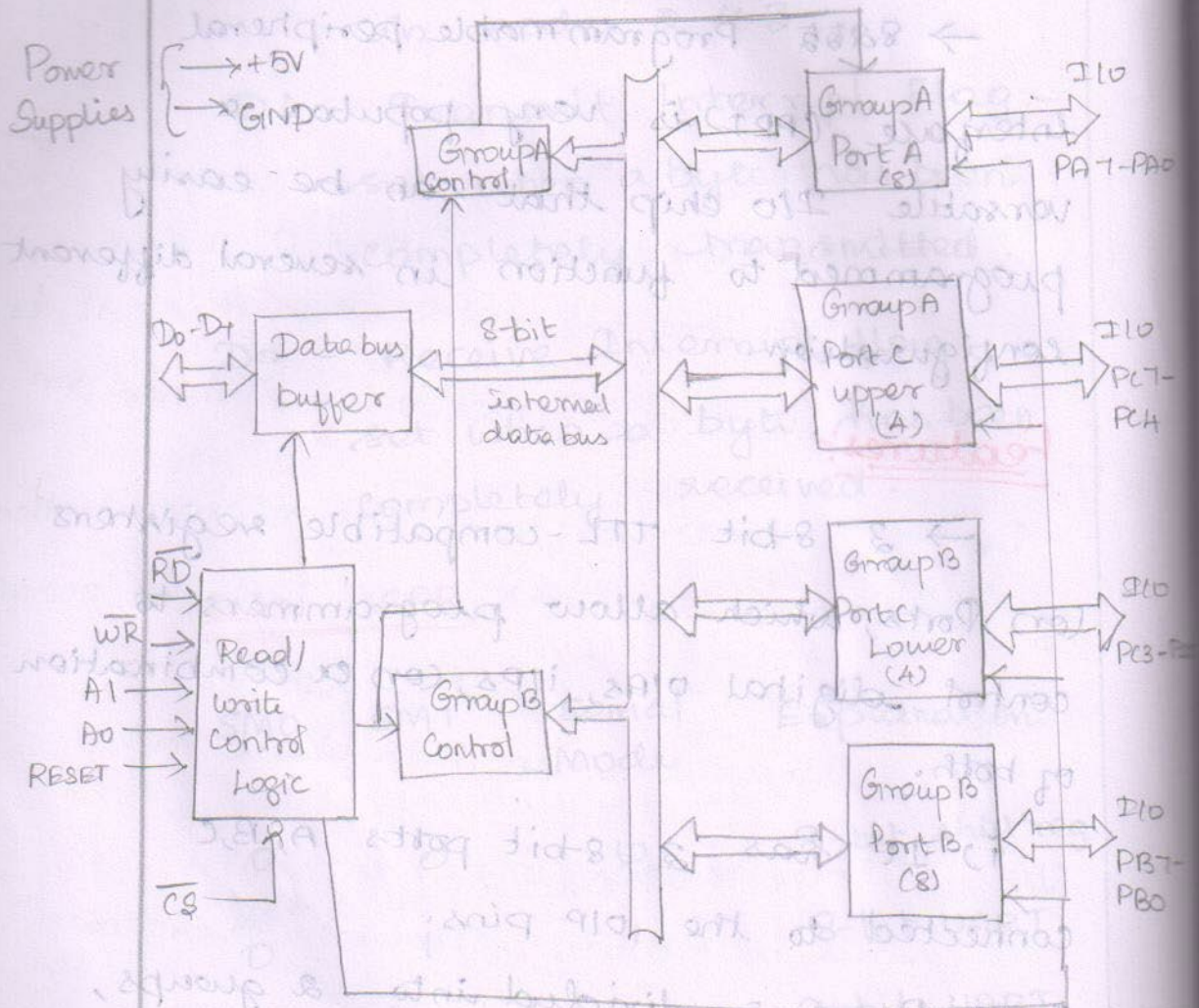
iii) All the ports can be programmed for simple I/O (or) handshake I/O in the Input/Output mode.

iv) Each port C bit can be set/reset individually in bit set/reset mode.

v) The bits of Port A & PCU are grouped as group A [GA].

vi) The bits of Port B & PCL are grouped as group B [GB].

Block diagram of Intel 8255:-



Address Lines & Register selection

in 8255:-

A1	A0	Registers selected
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Control Register

AD, AI are allotted for selecting one of the four registers available in 8255.

→ The control signals for reading from & writing into these registers are the active Low $\overline{RD}$ & $\overline{WR}$ signals.

→ The chip is selected by activating the active low chip select ($\overline{CS}$) signal.

→ The signal is obtained using decoder, which decodes the 8085 address lines & identifies 8255 address range.

→ A common reset signal such as RESET OUT of the 8085 processor can be applied to reset the 8255.

Operating Modes of 8255

→ **I/O Mode** $DT=1$ — i) mode 0

ii) mode 1

→ **BSR mode** $DT=0$

iii) mode 2

I/O Mode 1 operation:

→ provides a means for transferring I/O data to or from a specified port, in conjunction with strobes (or handshaking signals).

- Step 1:- The I/O device places data lines, i.e. Port A (or Port B). This is communicated

to 8255 by making the strobe iLP pin ($\overline{STB}$) low. $\overline{STB}$ is an active low signal applied through PC4 (or PC2).

Step 2:- The 8255 acknowledges the receipt of the data to the iLP device by making the iLP buffer full pin [$\overline{IBF}$] high. This also indicates that the data has been latched into the iLP port.

Step 3:- The 8255 then makes the interrupt request line [$\overline{INTR}$] high & applies interrupt to the processor. The signal is applied only when the interrupt enable signal [$\overline{INTE}$] is high.

PC2, PC4 can be controlled using BSR mode.

Step 4:-

→ processor reads the data from the corresponding iLP port.

Reading from the port is done by selecting the 8255 port & applying the active low $\overline{RD}$ signal.

Steps:-

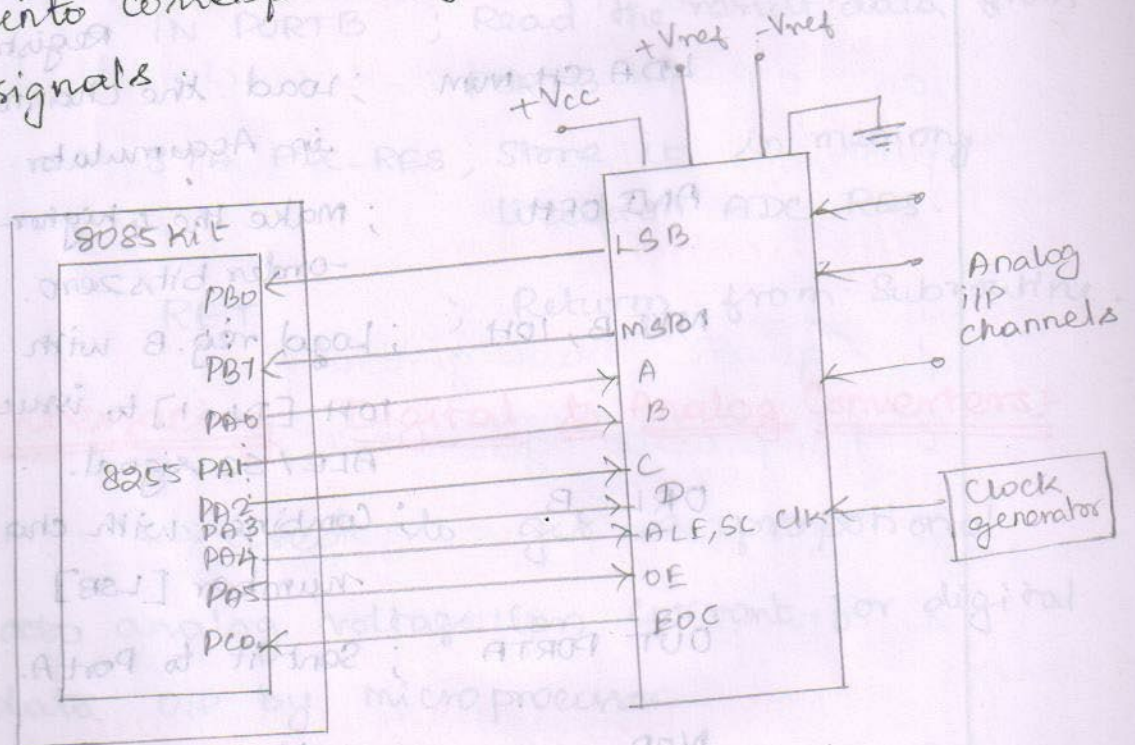
When $\overline{RD}$ - low, $INTR$ - Reset.

$\overline{DBF}$ is reset by rising edge of $\overline{RD}$ IIP.

$\therefore$ mode 1 allows IIP device to request service from the CPU by simply sending its data to the port & activating $\overline{STB}$ signal.

Interfacing Analog-to-Digital Converters.

$\Rightarrow$ Convert the IIP analog voltage level into corresponding Discrete Digital signals.



Interfacing ADC 0816 with 8255

$\Rightarrow$ ADC 0816 is an 8-bit successive approximation type ADC chip with a built-in analog multiplexer, which select one of 16 analog IIPs for conversion into digital format.

⇒ Select lines A, B, C & D are used to select one of 16 analog I/Ps IN0-IN15. The analog to digital conversion is started using the active HIGH control signal Start Conversion [SC].

Program For ADC conversion

Labels	Mnemonics	Comments
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ADCONVERT:	MVI A, 83H	; Load the 8255 Control word in Accumulator
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	OUT CONTROL-REG;	Move it to the Control Register
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	LDA CH-Num	; Load the channel no. in Accumulator.
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	ANI 0FH	; Make the 4 higher-order bits zero.
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	MVI B, 10H	; Load reg. B with 10H [D4=1] to issue ALE/SC signal.
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	DRL B	; Combine with channel number [LSB]
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	OUT PORTA	; Sent it to Port A.
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	NOP	; wait
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	NOP	; wait
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	MVI B, 00H	; Remove the ALE/SC signal
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	DRL B	; Combine
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	OUT PORT A	; sent it to Port B
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CHECK:	IN PORT C	; Read Port C to check for EOC.
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RAR ; Rotate the contents of the accumulator to bring EOC bit to the carry.

JNC CHECK ; If there is no EOC signal, repeat the process.

LDA CHNUM ; If EOC signal is received, load the channel number in Accumulator.

MVI B, 20H ; Make OE signal HIGH

ORL B ; Combine with channel number [LSB].

OUT PORTA ; Sent it to PORT A

IN PORTB ; Read the result data from PORTB.

STA ADC-RES ; Store it in memory location ADC-RES.

RET ; Return from Subroutine.

Interfacing Digital to Analog Converters:-

→ used to get a proportional analog voltage (or current) for digital data D/A by microprocessor.

→ The D/A converters are essential in micro-processor based systems as Real-world applications operate with analog data.

→ DAC - R-2R ladder Network

→ weighted Resistor Network

Eg:- 8-bit DAC can have 256 I/P combinations & so has a resolution of $\frac{1}{256}$ (or 0.39 % of Full scale O/P).

DAC 0800 is a common digital-to-analog converter chip that can be easily interfaced to 8085 through 8255. This section describes the interfacing of DAC 0800 with 8085 processor.

As DAC chips can be connected only to a Port, there is a need for an O/P port to be connected bt. the processor & DAC.

Applications:-

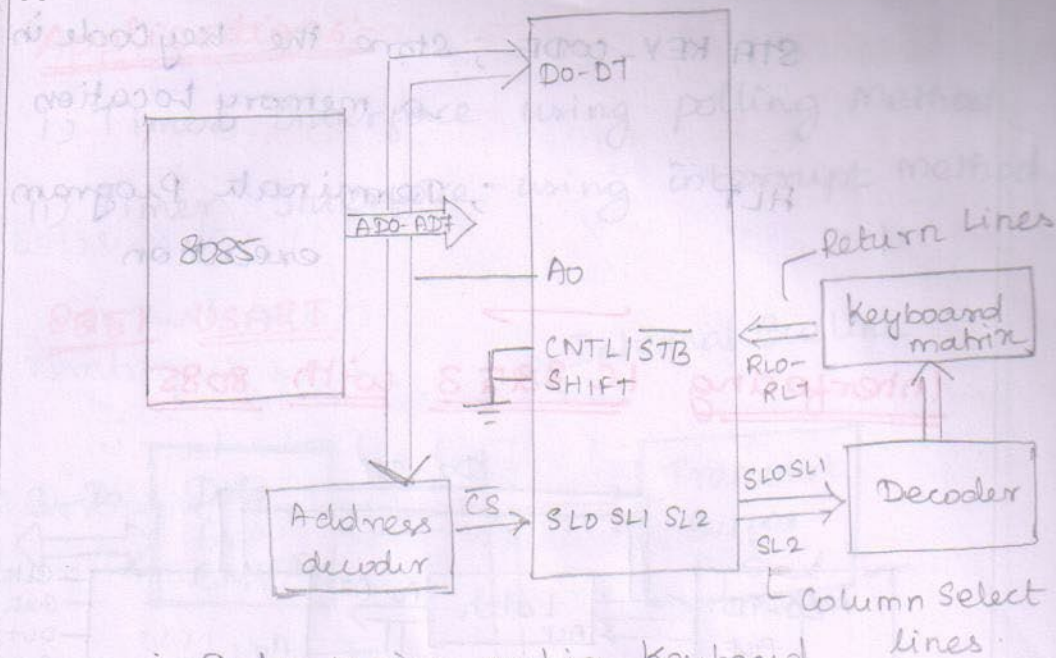
1. Square wave generation
2. Ramp wave generation
3. Stair Case wave generation
4. Sine wave generation.

Keyboard Interfacing using

8279

8279 plays 3 keyboard scan tasks,

- i) placing a Low in a Scan line
- ii) Checking for a Low on the return lines,
- iii) Detecting & de-bouncing the key is pressed.



i) Interfacing matrix Keyboard using IC 8279

Program for identifying a key press using IC 8279:

Labels Mnemonics Comments

START: MVI A, 00H ; Load the accumulator with mode set Command word

OUT COMMAND_PORT; O/P is to be Command port.

LOOP: IN COMMAND_PORT; Read the status word from the 8279.

ANI 0FH ; Mask the most significant 5 bits.

JZ LOOP ; If No Key is pressed loop again to read the status word.

MVI A, 50H ; If a Key is pressed, Load the read FIFO RAM Command in accumulator.

OUT COMMAND_PORT; O/P it to the Command PORT.

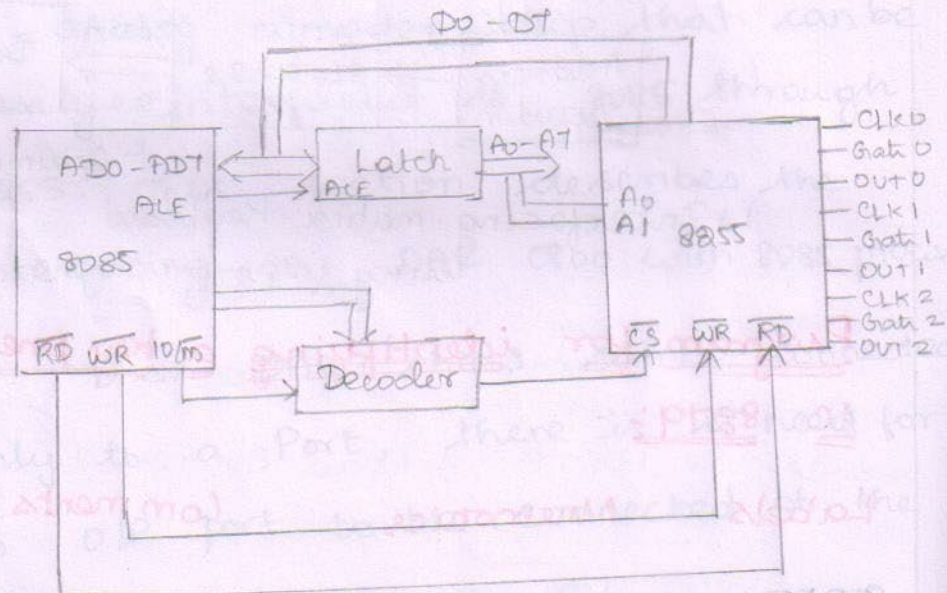
IN DATA_PORT ; Read FIFO RAM from Data PORT

STA KEY-CODE ; Store the key code in
a memory location

HLT

; Terminate Program
execution

Interfacing IC 8253 with 8085



→ The timer IC 8253 can be interfaced with 8085 as an I/O device.

→ In 8085-based sys, I/O devices requires an 8-bit address.

∴ 4 consecutive addresses (8-bit) are allotted to the 8253, for accessing the three counters & one control register.

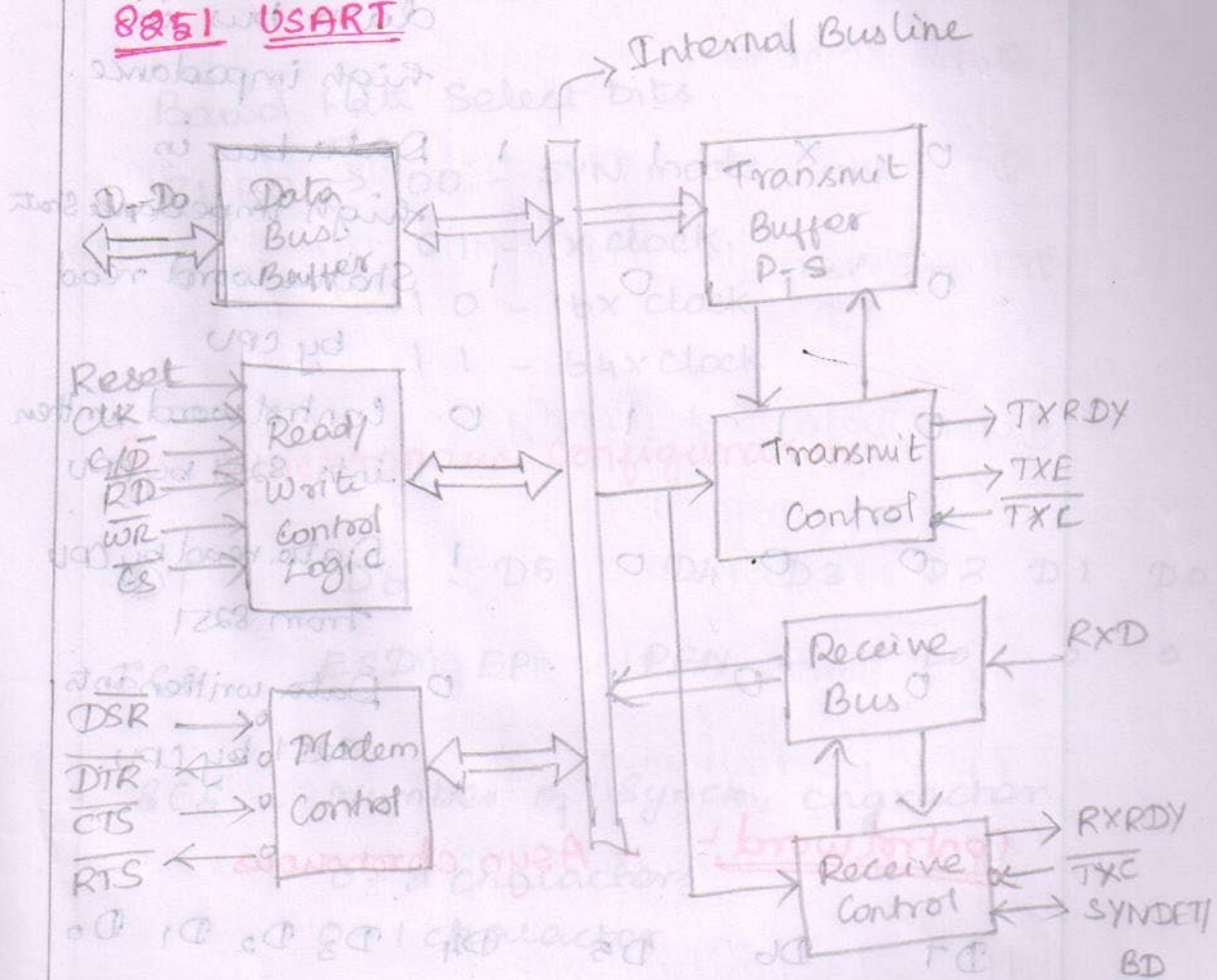
→ Lower order address bus & data bus must be de-multiplexed using a latch & ALE signal, like any 8085 system.

→ 8085 processor places the 8-bit I/O addresses on both higher-order & lower-order address

Applications:-

- i) Timer Interface using polling method
- ii) Timer Interface using interrupt method

8251 USART



→ Universal Synchronous, Asynchronous Receiver Transmitter used for serial data communication.

→ As peripheral device in a microcomputer system, 8251 receives parallel data from CPU & transmits them in serial data from outside, converts them into parallel data & sends them to CPU.

→ 8251 has 28 pins.

Basic Operations in 8251:-

$\overline{CS}$	$\overline{RD}$	$\overline{WR}$	Function
1	X	X	Chip not selected, data bus in high impedance.
0	X	1	Data bus in high impedance state
0	1	0	Status word read by CPU
0	1	1	Control word written into 8251 by CPU
0	0	0	Data read by CPU from 8251
0	0	1	Data written into 8251 by CPU.

Control word:- - Asynchronous

D_7 D_6 D_5 D_4 D_3 D_2 D_1 D_0
 S_1 S_0 EP PEN L L_0 $B1$ $B0$

Frame Control $\rightarrow$ Stop bit length.

$S_1 S_0$ - 00 - Inhibit

01 - 1 stop bit

10 - 1.5 stop bit

11 - 2 bits

Parity Check.

X_0 - Disable ; 01 - odd Parity

10 - Even parity.

$H, L_0 \rightarrow 00$ - 5 bits - character length
 01 - 6 bits
 10 - 7 bits
 11 - 8 bits

Band Rate Select bits.

$B1 B0 \rightarrow 00$ - SYN mode
 01 - 1x clock
 10 - 16x clock
 11 - 64x clock

For Synchronous configuration:-

D7	D6	D5	D4	D3	D2	D1	D0
SCS	ESD	EP	PEN	L1	L0	0	0

SCS - Number of Synchron. character
 0 - 2 characters
 0 - 1 character

ESD - Synchronous mode.
 0 - Internal Synchronisation
 1 - External Synchronisation.

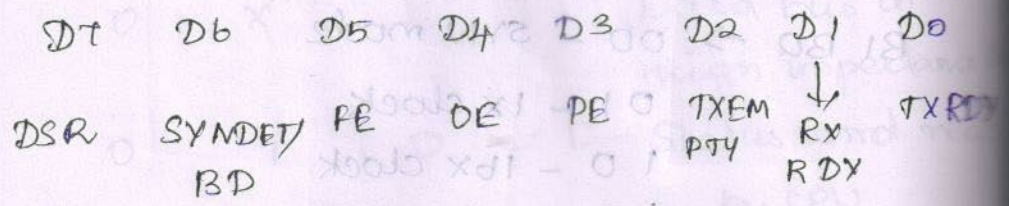
Parity check - 00 - Disable
 01 - odd parity
 10 - Even Parity

L1L0 - Char. Length.

00 - 5 bits
 01 - 6 bits
 10 - 7 bits
 11 - 8 bits

Status Word:

→ Possible to identify internal status of 8251 by reading a status word.



DSR - Data Set Ready

0 - DSR = 1

01 - DSR = 0

D6 - Same as the I/O pin

FE - 1 - Framing Error

DE - 1 - Overrun Error

PE - 1 - Parity Error

TXEMPTY - Same as in I/O Pin

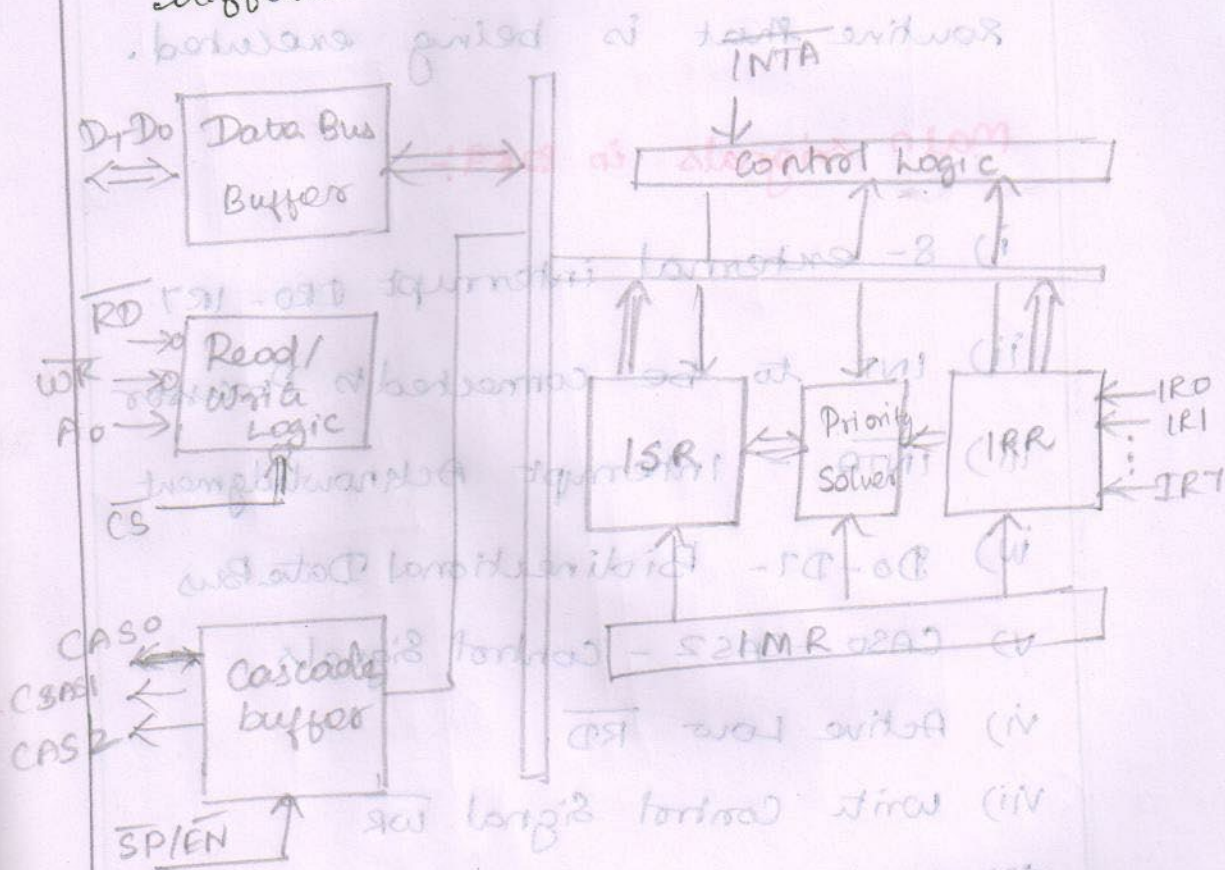
RXRDY - Same as in I/O Pin

TXRDY - Same as in I/O Pin

Interfacing with 8259

→ It supports 8 interrupts
i/p from the peripherals & issues
a single interrupt signal to the
processor.

- It supports cascading of 8259 ICs
- & multiplexes 64 interrupt sources into one
- It can set priorities for the interrupts
- mask the interrupt sources & provide different interrupt vectors



- i) IMR
- ii) ISR
- iii) IRR

IMR - Interrupt Mask Register
→ used to program the masking

of external interrupt sources.
This register is written into by programmer

IRR - Interrupt Request Register
→ used to store the interrupts

that have been sensed by 8259 at its I/P.

ISR - In service Register:-

→ maintains the list of interrupts that are currently being serviced & corresponding service routine that is being executed.

Main signals in 8259:-

- i) 8- external interrupt $IO/\overline{INT}$
- ii) $\overline{INT}$ to be connected to Processor
- iii) $\overline{INTA}$ - Interrupt Acknowledgment
- iv) D_0-D_7 - Bidirectional Data Bus
- v) CAS_0-CAS_2 - Control Signals
- vi) Active Low $\overline{RD}$
- vii) Write Control Signal $\overline{WR}$
- viii) A_0 to select control Registers
- ix) $\overline{CS}$ - chip Select
- x) $\overline{SP}/\overline{EN}$ - master or slave.

Don't mix up